

Session 14 (AIML) -Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.random.randint(10, 101, size=(5, 6))
4
5  print("Array:")
6  print(arr)
7
8  print("Shape:", arr.shape)
9  print("Size:", arr.size)
10 print("Data Type:", arr.dtype)
11
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Program File
ndled\libs\debugpy\launcher' '60502' '--' 'c:\Users\tanve\Desktop\ITR\Session 14.py'
Array:
[[65 74 54 44 61 96]
 [97 89 93 80 20 23]
 [69 32 89 81 74 19]
 [61 93 95 10 55 83]
 [82 31 20 86 56 52]]
Shape: (5, 6)
Size: 30
Data Type: int32
PS C:\Users\tanve\Desktop\ITR> 
```

Q2.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.random.randint(1, 51, 20)
4
5  print("Array:")
6  print(arr)
7
8  print("Minimum Value:", np.min(arr))
9  print("Index of Minimum:", np.argmin(arr))
10
11 print("Maximum Value:", np.max(arr))
12 print("Index of Maximum:", np.argmax(arr))
13
14 print("Sum:", np.sum(arr))
15 print("Mean:", np.mean(arr))
16 print("Standard Deviation:", np.std(arr))
```

OUTPUT:

```
ndled\libs\debugpy\launcher' '63992' '--' 'c:\Users\tanve\Desktop\ITR>
Minimum Value: 1
Index of Minimum: 19
Maximum Value: 48
Index of Maximum: 4
Sum: 405
Mean: 20.25
Standard Deviation: 16.893415877199022
PS C:\Users\tanve\Desktop\ITR>
```

Q3.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  matrix = np.random.randint(20, 81, size=(4, 5))
4
5  print("Matrix:")
6  print(matrix)
7
8  print("Minimum Value:", np.min(matrix))
9  print("Maximum Value:", np.max(matrix))
10
11 print("Sum:", np.sum(matrix))
12 print("Mean:", np.mean(matrix))
13 print("Standard Deviation:", np.std(matrix))
14
15 print("Row Sum:")
16 print(np.sum(matrix, axis=1))
17
18 print("Column Sum:")
19 print(np.sum(matrix, axis=0))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'
ndled\libs\debugpy\launcher' '61922' '--' 'c:\Users\tanve\Desktop\ITR'
Matrix:
[[58 71 22 61 26]
 [74 78 73 54 43]
 [67 76 43 40 70]
 [30 58 51 42 54]]
Minimum Value: 22
Maximum Value: 78
Sum: 1091
Mean: 54.55
Standard Deviation: 16.710700164864427
Row Sum:
[238 322 296 235]
Column Sum:
[229 283 189 197 193]
PS C:\Users\tanve\Desktop\ITR> 
```

Q4.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.arange(1, 25)
4
5  print("Original Array:")
6  print(arr)
7
8  a = arr.reshape(4, 6)
9  print("\nReshape (4,6)")
10 print(a)
11 print("Shape:", a.shape)
12
13 b = arr.reshape(3, 8)
14 print("\nReshape (3,8)")
15 print(b)
16 print("Shape:", b.shape)
17
18 c = arr.reshape(2, 3, 4)
19 print("\nReshape (2,3,4)")
20 print(c)
21 print("Shape:", c.shape)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:: cd 'c:\Users\tanve\Desktop\ITR'; & 'C:
ndled\libs\debugpy\launcher' '53509' '--' 'c:\Users\tanve\Desktop\ITR\Sess
Original Array:
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24]

Reshape (4,6)
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]
 [13 14 15 16 17 18]
 [19 20 21 22 23 24]]
Shape: (4, 6)

Reshape (3,8)
[[ 1  2  3  4  5  6  7  8]
 [ 9 10 11 12 13 14 15 16]
 [17 18 19 20 21 22 23 24]]
Shape: (3, 8)

Reshape (2,3,4)
[[[ 1  2  3  4]
   [ 5  6  7  8]
   [ 9 10 11 12]]
 [[13 14 15 16]
   [17 18 19 20]
   [21 22 23 24]]]
Shape: (2, 3, 4)
PS C:\Users\tanve\Desktop\ITR> █
```

Q5.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.array([[10,20,30,40],
4                 [50,60,70,80],
5                 [90,100,110,120]])
6
7  print("Array:")
8  print(arr)
9
10 print("\nFirst Row:")
11 print(arr[0])
12
13 print("\nLast Column:")
14 print(arr[:, -1])
15
16 print("\nCenter 2x2 Submatrix:")
17 print(arr[1:3, 1:3])
18
19 print("\nEven Numbers:")
20 print(arr[arr % 2 == 0])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'
ndled\libs\debugpy\launcher' '60707' '--' 'c:\Users\tanve\Desktop\ITR'
Array:
[[ 10  20  30  40]
 [ 50  60  70  80]
 [ 90 100 110 120]]

First Row:
[10 20 30 40]

Last Column:
[ 40  80 120]

Center 2x2 Submatrix:
[[ 60  70]
 [100 110]]

Even Numbers:
[ 10  20  30  40  50  60  70  80  90 100 110 120]
PS C:\Users\tanve\Desktop\ITR>
```

Q6.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.random.randint(1, 101, size=(5,5))
4
5  print("Original Matrix:")
6  print(arr)
7
8  print("\nDiagonal Elements:")
9  print(np.diag(arr))
10
11 print("\nElements Greater Than 50:")
12 print(arr[arr > 50])
13
14 arr[arr < 30] = 0
15
16 print("\nModified Matrix:")
17 print(arr)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\U
ndled\libs\debugpy\launcher' '49697' '--' 'c
Original Matrix:
[[64  7 13 62 12]
 [74 56 87 87 45]
 [64 74 78 30 18]
 [39 45 73  4 25]
 [61 92 14 45 89]]

Diagonal Elements:
[64 56 78  4 89]

Elements Greater Than 50:
[64 62 74 56 87 87 64 74 78 73 61 92 89]

Modified Matrix:
[[64  0  0 62  0]
 [74 56 87 87 45]
 [64 74 78 30  0]
 [39 45 73  0  0]
 [61 92  0 45 89]]
```


Q7.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  a = np.array([10,20,30,40])
4  b = np.array([1,2,3,4])
5
6  print("Addition:")
7  print(a + b)
8
9  print("\nSubtraction:")
10 print(a - b)
11
12 print("\nMultiplication:")
13 print(a * b)
14
15 print("\nDivision:")
16 print(a / b)
17
18 print("\nPower:")
19 print(a ** b)
20
21 print("\nModulo:")
22 print(a % b)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\ndled\libs\debugpy\launcher' '63297' '--'
Addition:
[11 22 33 44]

Subtraction:
[ 9 18 27 36]

Multiplication:
[ 10 40 90 160]

Division:
[10. 10. 10. 10.]

Power:
[ 10 400 27000 2560000]

Modulo:
[0 0 0 0]
PS C:\Users\tanve\Desktop\ITR> 
```

Q8.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  A = np.array([[1,2,3],
4               [4,5,6],
5               [7,8,9]])
6
7  B = np.array([[9,8,7],
8               [6,5,4],
9               [3,2,1]])
10
11 print("Element-wise Multiplication:")
12 print(A * B)
13
14 print("\nMatrix Multiplication:")
15 print(A @ B)
16
17 print("\nDifference:")
18 print("* -> Element-wise multiplication")
19 print("@ -> Matrix multiplication")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd '
ndled\libs\debugpy\launcher' '54073' '--
Element-wise Multiplication:
[[ 9 16 21]
 [24 25 24]
 [21 16  9]]

Matrix Multiplication:
[[ 30  24  18]
 [ 84  69  54]
 [138 114  90]]

Difference:
* -> Element-wise multiplication
@ -> Matrix multiplication
PS C:\Users\tanve\Desktop\ITR> 
```


Q9.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  arr = np.random.randn(6,6)
4
5  print("Matrix:")
6  print(arr)
7
8  print("Shape:", arr.shape)
9  print("Size:", arr.size)
10 print("Data Type:", arr.dtype)
11
12 print("\nMaximum Index:")
13 print(np.unravel_index(np.argmax(arr), arr.shape))
14
15 print("Minimum Index:")
16 print(np.unravel_index(np.argmin(arr), arr.shape))
17
18 print("\nTop-left 3x3 Matrix:")
19 print(arr[:3, :3])
20
21 arr = np.abs(arr)
22
23 print("\nModified Matrix:")
24 print(arr)
25
26 print("Mean:", np.mean(arr))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Program F
'c:\Users\tanve\Desktop\ITR\Session 14.py'
Matrix:
[[ 8.41377212e-01 -6.01026718e-01 1.00233674e-01 -2.46729636e-02
 -1.49557986e+00 -8.26585349e-01]
 [-7.85816078e-01 -8.50244913e-01 -4.77583982e-01 1.19119687e+00
 -1.40970831e+00 -4.93446517e-01]
 [-5.50804700e-01 -9.51870703e-01 5.46191770e-01 -1.06823849e+00
 -8.45166821e-02 -6.28568516e-01]
 [ 5.20489171e-01 -4.92304750e-03 -1.12089695e-03 -1.28441970e+00
 -2.96316931e-01 5.22832086e-01]
 [ 9.02176392e-01 1.81761594e+00 1.90330589e+00 8.42104104e-01
 -3.90409359e-01 -1.39681231e+00]
 [-7.23593957e-01 4.88262335e-01 -1.35860336e+00 9.16483895e-02
 -1.58185287e+00 1.58215354e+00]]
Shape: (6, 6)
Size: 36
Data Type: float64

Maximum Index:
(4, 2)
Minimum Index:
(5, 4)

Top-left 3x3 Matrix:
[[ 0.84137721 -0.60102672 0.10023367]
 [-0.78581608 -0.85024491 -0.47758398]
 [-0.5508047 -0.9518707 0.54619177]]

Modified Matrix:
[[8.41377212e-01 6.01026718e-01 1.00233674e-01 2.46729636e-02
 1.49557986e+00 8.26585349e-01]
 [7.85816078e-01 8.50244913e-01 4.77583982e-01 1.19119687e+00
 1.40970831e+00 4.93446517e-01]
 [5.50804700e-01 9.51870703e-01 5.46191770e-01 1.06823849e+00
 8.45166821e-02 6.28568516e-01]
 [5.20489171e-01 4.92304750e-03 1.12089695e-03 1.28441970e+00
 2.96316931e-01 5.22832086e-01]
 [9.02176392e-01 1.81761594e+00 1.90330589e+00 8.42104104e-01
 3.90409359e-01 1.39681231e+00]
 [7.23593957e-01 4.88262335e-01 1.35860336e+00 9.16483895e-02
 1.58185287e+00 1.58215354e+00]]
Mean: 0.7954528774343829
PS C:\Users\tanve\Desktop\ITR> 
```

Q10.

PROGRAM:

```
Session 14.py > ...
1  import numpy as np
2
3  marks = np.random.randint(30, 101, size=(10,5))
4
5  print("Student Marks:")
6  print(marks)
7
8  total = np.sum(marks, axis=1)
9  average = np.mean(marks, axis=1)
10
11 print("\nTotal Marks:")
12 print(total)
13
14 print("\nAverage Marks:")
15 print(average)
16
17 highest = np.argmax(total)
18 lowest = np.argmin(total)
19
20 print("\nHighest Scoring Student:", highest + 1)
21 print("Total:", total[highest])
22
23 print("\nLowest Scoring Student:", lowest + 1)
24 print("Total:", total[lowest])
25
26 print("\nClass Mean:", np.mean(marks))
27 print("Class Standard Deviation:", np.std(marks))
28
29 print("\nMarks of Top 3 Students:")
30 print(marks[:3])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:: cd 'c:\Users\tanve\Desktop\ITR';  
n 14.py'  
Student Marks:  
[[ 51 60 30 37 83]  
 [ 65 64 98 81 46]  
 [100 51 72 49 99]  
 [ 98 31 38 42 35]  
 [ 84 51 98 57 76]  
 [ 97 56 95 49 86]  
 [ 47 58 96 53 93]  
 [ 88 39 59 37 89]  
 [ 42 81 63 59 65]  
 [ 89 78 100 33 34]]  
  
Total Marks:  
[261 354 371 244 366 383 347 312 310 334]  
  
Average Marks:  
[52.2 70.8 74.2 48.8 73.2 76.6 69.4 62.4 62. 66.8]  
  
Highest Scoring Student: 6  
Total: 383  
  
Lowest Scoring Student: 4  
Total: 244  
  
Class Mean: 65.64  
Class Standard Deviation: 22.672238530855306  
  
Marks of Top 3 Students:  
[[ 51 60 30 37 83]  
 [ 65 64 98 81 46]  
 [100 51 72 49 99]]  
PS C:\Users\tanve\Desktop\ITR> 
```